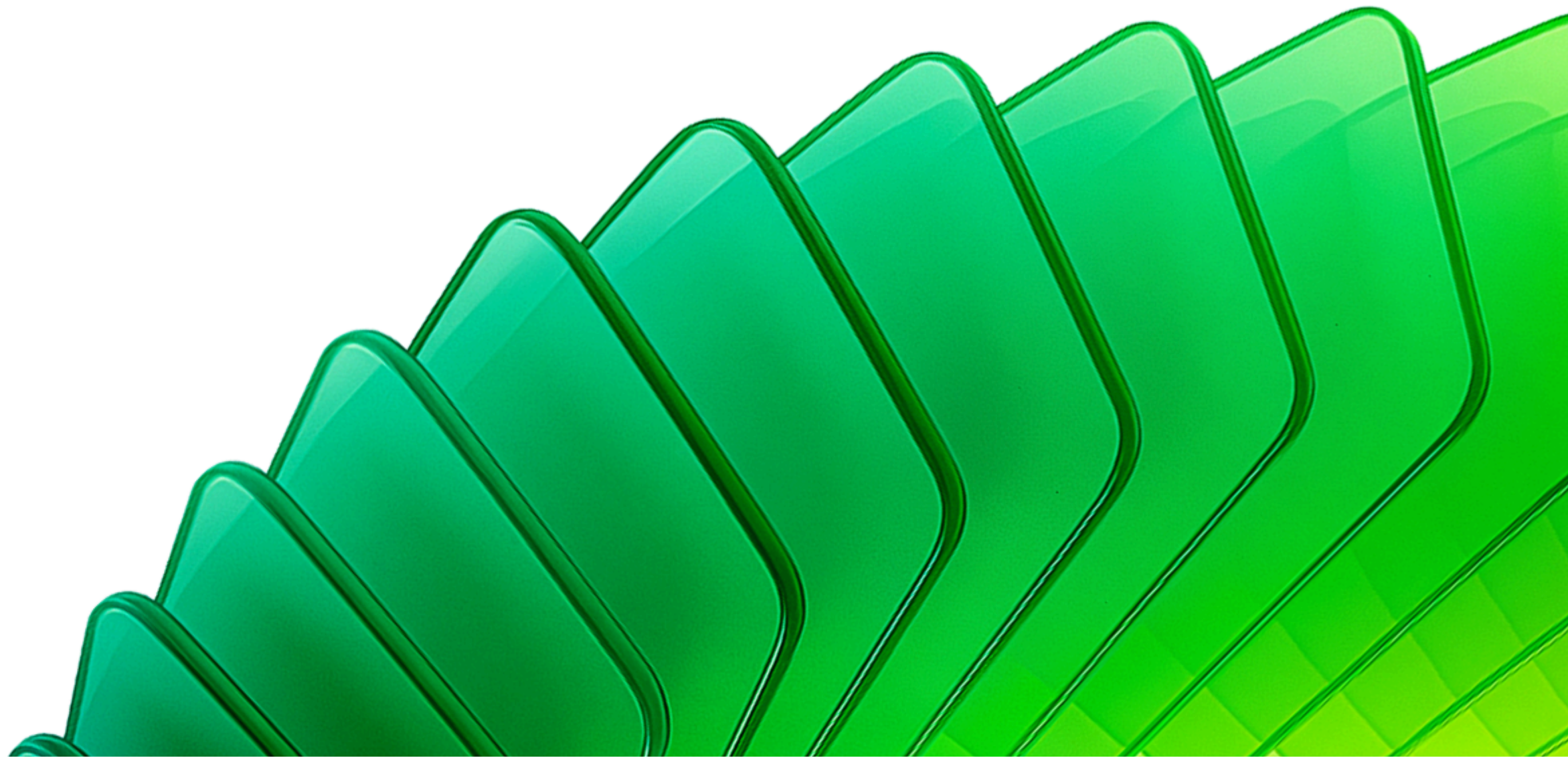


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PARTICIPANT DETAILS

Participant Name - Hrituraj Saha.

Problem Statement - Build and deploy a single AI agent using ADK and Gemini that is hosted on Cloud Run and performs one clearly defined task.

BRIEF ABOUT THE IDEA

The idea behind the **Student Guide Agent** is to act like a smart, always-available tutor that helps students understand any topic in a clear and structured way. Instead of just giving raw answers, the system is designed to teach. When a student asks a question, anything from “What is Machine Learning?” to “Explain Newton’s Laws” the agent first understands the query, gathers accurate information (using tools like Wikipedia if needed), and then breaks it down into simple, easy-to-understand concepts.

What makes it useful is how it presents the information. Rather than long paragraphs, it organizes everything into clean study notes like definition, key points, examples, and a short summary. This makes it much easier for students to revise, remember, and even use it directly for exams. In simple terms, it’s like having a personal tutor + note-maker combined into one system. It doesn’t just answer questions, it helps you actually learn and retain the concept.

The bigger vision is to expand this into a complete learning assistant that can generate quizzes, adapt to a student’s level, and even guide them step by step through complex topics.

MEETING THE BUILD CRITERIA

Approach

Built a Student Guide Agent (ADK) that converts queries → structured study notes.

Capability

Single task - Question → Explanation (Definition, Points, Examples, Summary).

Gemini

Used for understanding queries & generating clear responses.

Deployment

Deployed on Cloud Run as an HTTP API.

Alignment

- ADK used.

- Gemini used.

- One clear task.

- HTTP input/output.

OPPORTUNITIES

Differentiation

- Not just Q&A → delivers structured study notes.
- Focused on learning clarity, not raw answers.

Problem Solving

- Simplifies complex topics into easy, exam-ready content.
- Saves time by combining explanation + notes in one step.

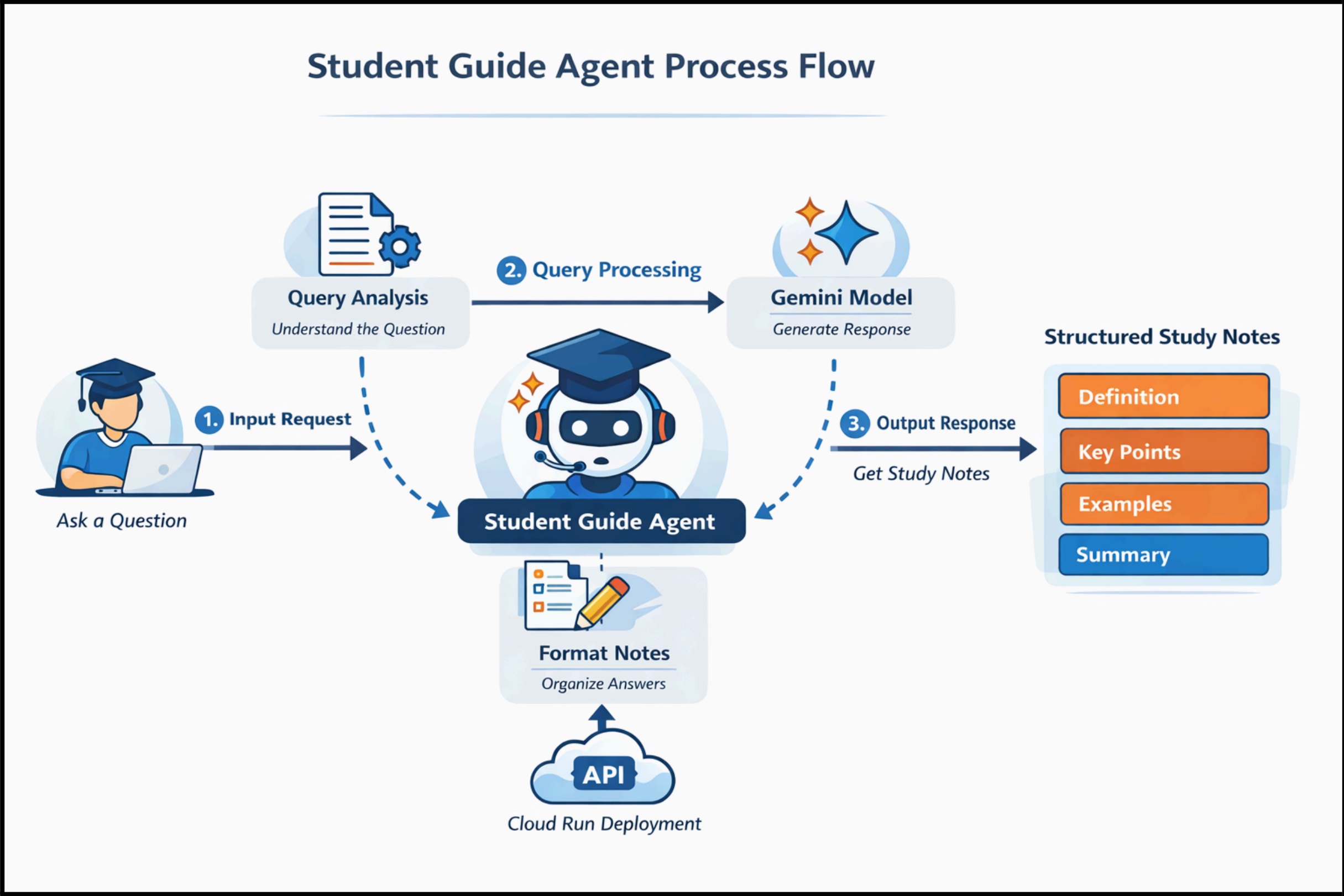
USP (Unique Value)

- Personal tutor + note-maker in one agent.
- Clean, consistent format (Definition, Points, Examples, Summary).
- Lightweight, API-based → easy to integrate anywhere.

LIST OF FEATURES OFFERED BY THE SOLUTION

- Accepts natural language queries.
- Uses Gemini for intelligent responses.
- Generates structured study notes.
- Provides clear concept explanations.
- Delivers quick, concise answers.
- Maintains consistent output format.
- Accessible via HTTP API (Cloud Run).
- Built with modular ADK architecture.
- Supports external knowledge lookup (Wikipedia).

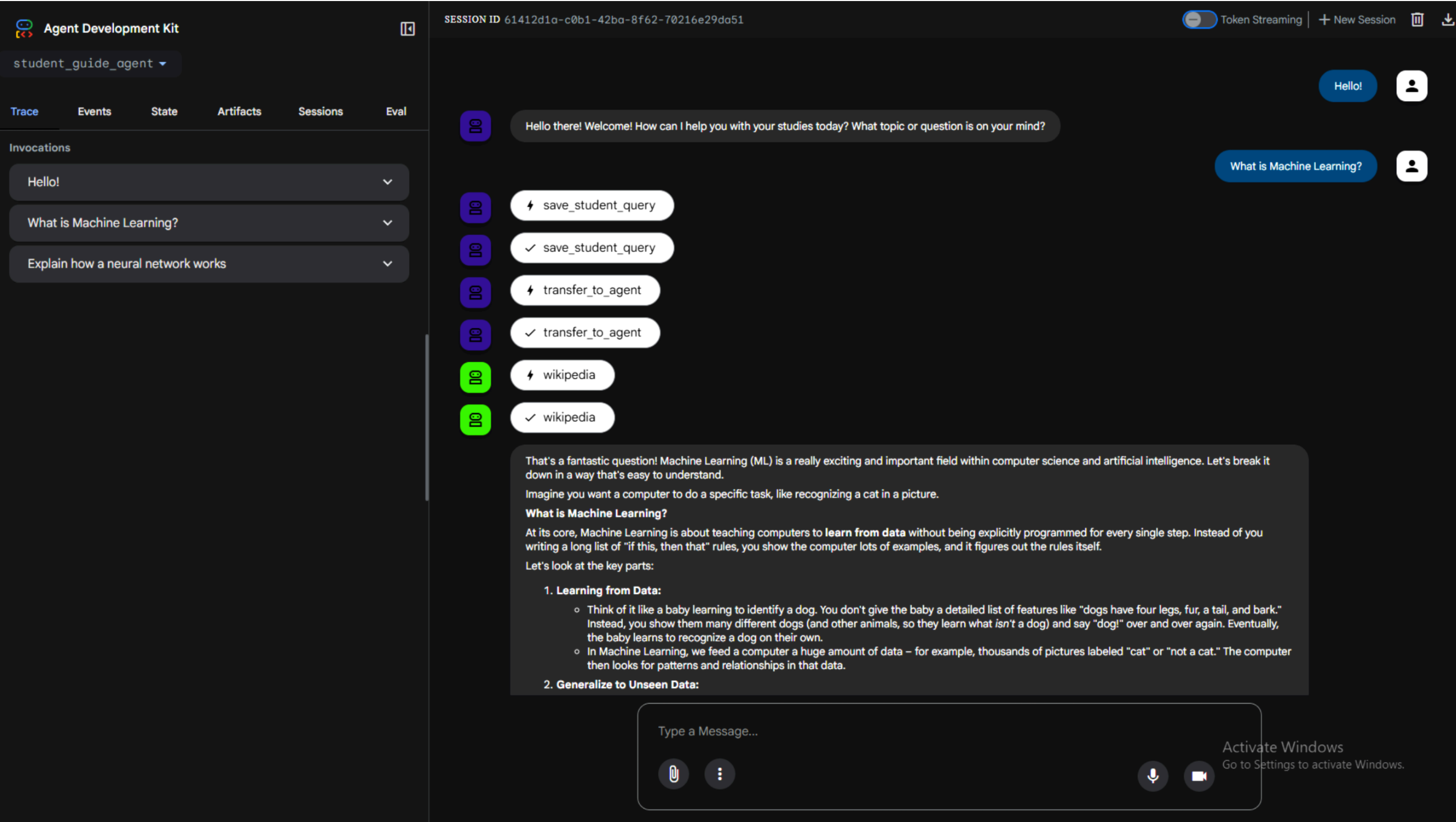
PROCESS FLOW DIAGRAM OR USE-CASE DIAGRAM



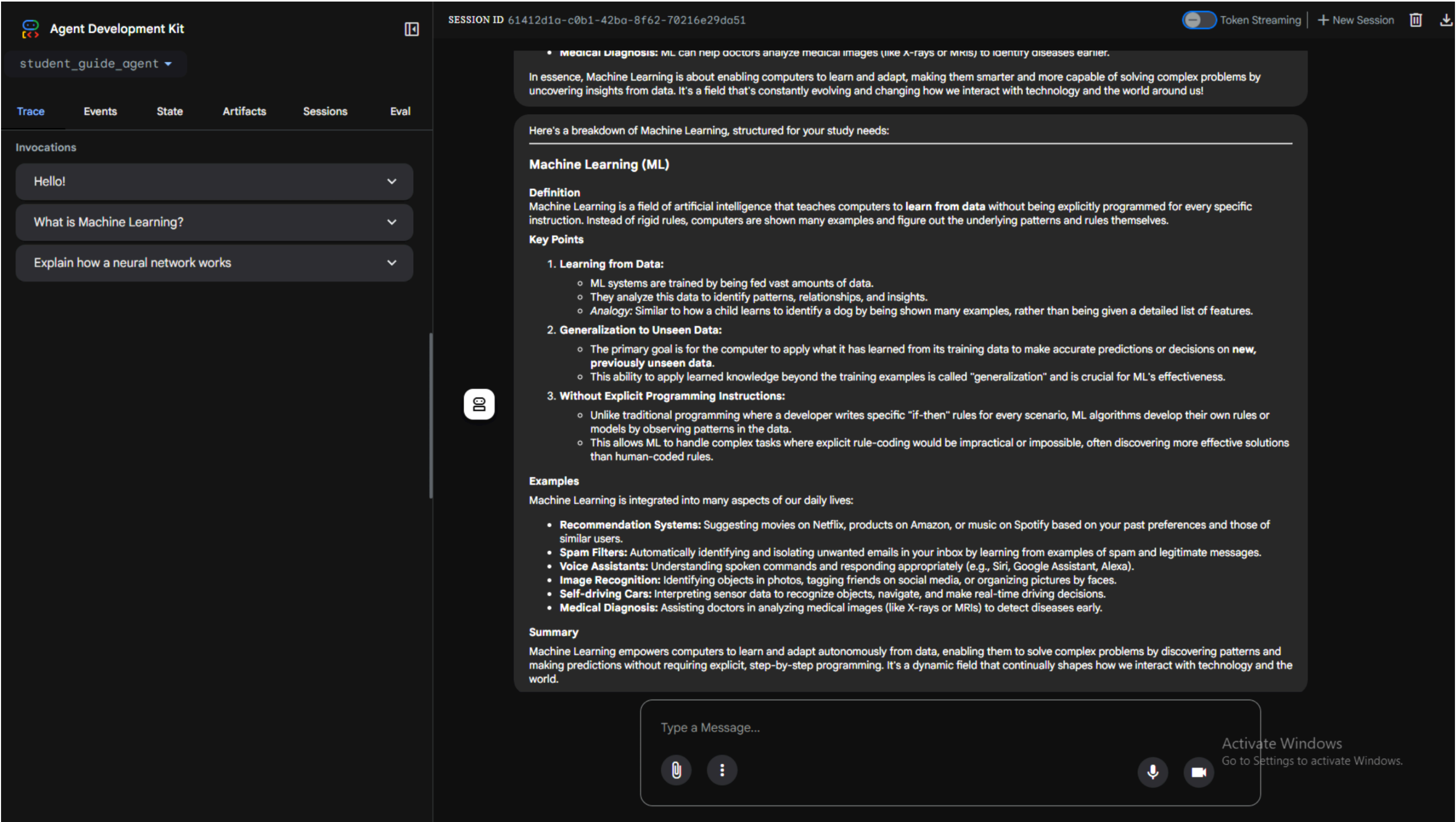
TECHNOLOGIES TO BE USED IN THE SOLUTION

- Google ADK – Agent orchestration and workflow design.
- Gemini Model – Core AI for understanding & generating responses.
- Google Cloud Run – Deployment and hosting.
- HTTP API – Request handling and endpoint exposure.
- Wikipedia API – Fetching factual information.
- Python – Core programming language.
- Google Cloud Logging – Monitoring and debugging.
- Docker – Containerization for Cloud Run deployment.

SNAPSHOTS OF THE PROTOTYPE



SNAPSHOTS OF THE PROTOTYPE





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Thank you

